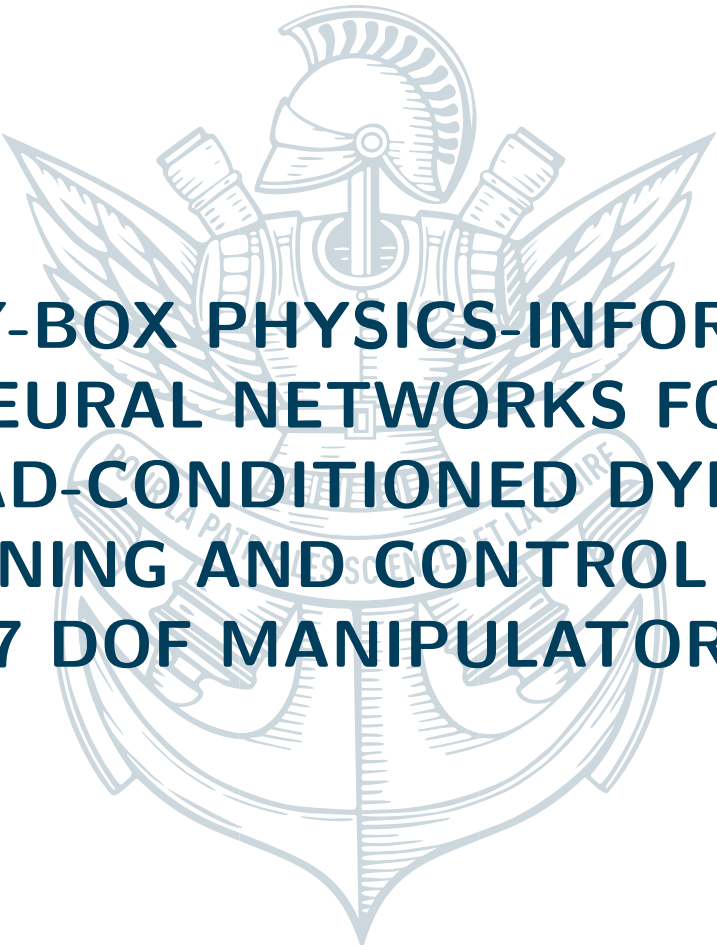




GREY-BOX PHYSICS-INFORMED NEURAL NETWORKS FOR PAYLOAD-CONDITIONED DYNAMICS LEARNING AND CONTROL OF A 7 DOF MANIPULATOR

**An automated pipeline from URDF file to controlled robot,
and 1 kHz real-time control of a simulated Franka Emika
Panda (MuJoCo)**

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Abstract. Torque-level manipulator control requires a faithful dynamics model, which the analytical rigid-body model derived from a URDF does not provide once friction and transmission compliance are accounted for. No published work yet composes an analytical term with a learned residual simultaneously across seven axes, under hard physical constraints, payload-conditioned, and queried at 1 kHz. We build an automated pipeline taking the URDF file as sole input and producing a grey-box model $\tau_{\text{pred}} = \tau_{\text{rbd}} + \tau_{\text{res}}$, whose analytical term is evaluated by RNEA with no manual derivation and whose payload-conditioned neural residual exposes three increasing degrees of guarantee (unconstrained, imposed by an augmented Lagrangian, structurally guaranteed by construction). The model feeds a 1,000 Hz computed-torque-plus-PD law whose gains derive from an error bound measured on a held-out test set. Over 14,830 held-out samples the grey-box model reaches 0.29 N m to 1.02 N m per joint (0.64 % to 3.09 % of the torque limit) against 1.46 N m mean for the analytical model alone, with zero torque-limit violations where the analytical model violates them on 0.99 % of samples. Two results run against the starting hypothesis and are reported as such: a black box of identical capacity and budget does *better* than the grey box (0.45 N m against 0.62 N m mean), and the learned residual *degrades* the joints the analytical term already models well — by a factor of 6.9 on joint 7. The same phenomenon appears in closed loop: disabling the residual divides joint 5’s steady-state bias by 22 and halves the Cartesian flange error.

CONTENTS

1	Introduction	4
1.1	A model that is exact and nevertheless wrong	4
1.2	Objective and scientific lock	4
1.3	Contributions and state of progress	5
2	Related work	5
3	Preliminaries	6
4	Problem formulation	7
5	Method	7
5.1	Decomposition: what is computed, what is learned	7
5.2	The learned residual and its three degrees of guarantee	8
5.3	Identification and data hygiene	9
5.4	Closing the loop	9
6	Validation and results	10
6.1	Baselines and comparison protocol	10
6.2	A caveat on the partition, and on what it distorts	12
6.3	Accuracy of the learned model	13
6.4	Ablations: what each component actually contributes	13
6.5	Compliance with the physical guarantees	14
6.6	Closed-loop trajectory tracking	14
6.7	Ablation of the learned residual under cross-simulator transfer	14
7	Discussion and limitations	15
7.1	The uncorrected optimisation bias	15
7.2	Limitations of the integration with the physics engine	15
7.3	Structural and methodological limitations	16
8	Conclusion	17

1

INTRODUCTION

1.1 A MODEL THAT IS EXACT AND NEVERTHELESS WRONG

This work was carried out as part of a Master’s research internship at the University of Melbourne, within the Human-Computer-Interaction team, one of whose research directions is human-agent interaction. A robot that shares a workspace with a person, or that responds to a high-level instruction rather than to a programmed trajectory, can only be safe and reactive if its internal dynamic model is both accurate and queryable in real time. It is this double requirement model accuracy, availability inside the control loop that motivates choosing a physically instrumentable manipulator such as the Franka Emika Panda as the target of this work, beyond the interest of the learning method itself.

Precise control of a manipulator arm relies on knowledge of its dynamics, that is, of the mapping that relates a desired joint trajectory to the motor torques that realise it. This relation admits a closed analytical form, recalled in section 3, entirely determined once the geometry and the inertial parameters of the robot are known and therefore computable directly from a URDF file (*Unified Robot Description Format*).

The problem is that this analytical form is wrong on a real robot. Dry and viscous friction in the harmonic drives, transmission compliance, mechanical backlash, temperature-dependent losses and actuator wear produce a systematic gap between the torque predicted by the rigid-body model and the torque actually measured. This gap, called the residual torque here, is of the order of several newton-metres on a Franka Emika Panda and cannot be modelled analytically in a reliable way. This is precisely where machine learning comes in and it is also where the entire scientific content of the present work lies: not in replacing the rigid-body model, but in handling what it does not say.

Two routes oppose each other. The first learns the dynamics end to end, as a black box: a neural network receives the state and predicts the torque. This approach is flexible but data-hungry, guarantees no physical property and generalises poorly outside the training distribution. The second, called *physics-informed*, constrains the network to respect the structure of the equations of mechanics either by writing that structure into the architecture, or by imposing it through loss terms. Deep Lagrangian networks [6] and, more recently, the work of Liu, Borja and Della Santina [5] have established the theoretical and experimental viability of this second route on real manipulators.

1.2 OBJECTIVE AND SCIENTIFIC LOCK

The objective of this work is to **build and validate an automatic pipeline that takes as input the description file of a manipulator alone and produces as output a robot effectively controlled** by a learned dynamic model, queried at every period of the control loop. This pipeline comprises the learning of a grey-box dynamic model, its deployment in a real-time control law, and a manipulation task serving as an integration test.

Stated this way, the objective may look like engineering. Three distinct difficulties make it non-trivial, and they are what defines the contribution.

Generality is obtained at the expense of accuracy. The literature obtains its best results by deriving the equations of one particular robot by hand, then fitting a network on top of that structure. Making the procedure automatic from a plain URDF means giving up any manual derivation: the kinematic graph, the network dimensions, the excitation of the training trajectories and the physical bounds must all be extracted from the description file.

Physical guarantees survive learning badly. A freely learned residual can inject energy into the system physically absurd for a term meant to represent friction and predict torques above the actuator limits. On a 7-degree-of-freedom system these constraints must be satisfied simultaneously on seven joints whose torque limits differ by a factor of 7 (87 N m for the first four joints, 12 N m for the last three), which naturally unbalances the optimisation.

Offline accuracy does not make a controller. A low prediction error on a test set says nothing about closed-loop stability. Inserting a neural network into a 1,000 Hz loop imposes sub-millisecond inference, reliable real-time scheduling, and a control law whose error can be bounded.

1.3 CONTRIBUTIONS AND STATE OF PROGRESS

This work positions itself explicitly with respect to the state of the art defined by Liu *et al.* [5], the closest reference: the same robot (Franka Panda 7 DoF), the same class of methods (physics-informed networks), the same purpose (model, then control). Four contributions are claimed.

1. **An automated “URDF to model” pipeline.** The URDF file is the sole input: the kinematic graph and the inertial parameters are extracted from it by Pinocchio, and the analytical term is evaluated by RNEA with no manual derivation whatsoever.
2. **The integration of the learned model into a 1,000 Hz control loop.** The model is queried at every control period by a ROS2 node that drives a Franka simulated under MuJoCo.
3. **The scaling up to 7 DoF under hard physical constraints.** The torque-limit and dissipativity constraints are imposed simultaneously by an augmented Lagrangian with dual ascent, and completed by a structural guarantee of dissipativity.
4. **A measurement of the cost of cross-simulator transfer.** A controlled ablation of the residual in closed loop (section 6.7) quantifies what it costs to deploy, under MuJoCo, a residual learned on MuJoCo. It provides the baseline that the frozen-backbone fine-tuning protocol implemented but not yet validated has to beat.

The actual state of progress must be stated up front: contributions 1 and 3 are implemented and quantitatively validated; contribution 2 is validated in closed loop, but the inference latency has not been formally measured on a dedicated bench; contribution 4 is measured, but its corollary does fine-tuning actually reduce the gap? remains to be validated (section 7.3).

2 RELATED WORK

The axis of reading adopted here is the one that genuinely separates the methods from one another: **where the physical knowledge is injected into the learning problem, and what it guarantees in return.** The four families below are ordered from the strongest constraint to the weakest.

Parameter identification. Keep the analytical structure (1) intact and seek only to identify its parameters. Sutanto *et al.* [7] propose the modern, differentiable version of this. The benefit is interpretability; the limitation is structural: a rigid model can represent neither nonlinear friction nor compliance, whatever the quality of the identification.

Physics written into the architecture. Move the physics into the architecture of the network. Deep Lagrangian networks [6] learn the inertia matrix in Cholesky-factorised form, which guarantees its positive-definiteness by construction. Duong *et al.* [3] transpose the idea into the port-Hamiltonian formalism on Lie groups. The guarantee then does not depend on the data, but the price paid is the loss of automatic generality: quantities that the URDF already provides exactly (inertia, gravity) have to be relearned. Moreover, although the port-Hamiltonian formalism intrinsically guarantees total passivity, it turns out to be disproportionate when the only targeted property is the non-injection of energy by the dissipative terms.

Grey-box models: composing analytical and residual. Compose a known analytical term with a learned residual. Djeumou *et al.* [2] give the reference formulation. Deng *et al.* [1] usefully warn that a model can display an excellent mean error while producing absurd maximum errors — a direct argument in favour of hard constraints rather than a plain least-squares minimisation. The architecture developed in section 5 is an instance of this grey family.

Constraints specific to the closed loop. The closed loop imposes its own requirements, often absent from purely offline work. The control rate disqualifies heavy integrations such as extended Kalman filtering [8] at 1,000 Hz. Hu *et al.* [4] establish a fact that is crucial for payload conditioning: the backlash error depends on position and velocity, but not on the carried mass — which justifies conditioning only the residual term on δ , rather than the whole model.

The lock addressed here. The grey composition is established. It has not yet been carried *simultaneously* to seven axes, under hard physical constraints, payload-conditioned, and queried at 1,000 Hz inside a real control loop. It is the conjunction of these four requirements, and not each of them taken in isolation, that defines the contribution of the present work.

3 PRELIMINARIES

Notation. $\mathbb{R}^n$ denotes the n -dimensional Euclidean space, with $n = 7$ for the Franka Emika Panda considered throughout this document. The vectors $q, \dot{q}, \ddot{q} \in \mathbb{R}^n$ denote joint positions, velocities and accelerations respectively. The scalar $\delta \in \mathbb{R}_+$ denotes the mass carried at the end effector (the payload). The vector $\bar{\tau} \in \mathbb{R}_+^n$ collects the per-joint torque limits, and $\bar{\tau}_j$ denotes its j -th component. For a vector v , $|v|$ is understood component-wise.

Rigid-body dynamic model. For a rigid serial robot with n degrees of freedom, the inverse dynamic model takes the classical form

$$\tau = M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + G(q), \quad (1)$$

where $M(q) \in \mathbb{R}^{n \times n}$ is the inertia matrix (symmetric positive definite), $C(q, \dot{q}) \in \mathbb{R}^{n \times n}$ the matrix of Coriolis and centrifugal terms, $G(q) \in \mathbb{R}^n$ the gravity vector and $\tau \in \mathbb{R}^n$ the vector of joint torques. Equation (1) is entirely determined once the geometry and the inertial parameters of the robot are known; it is computed efficiently by the recursive Newton–Euler algorithm (RNEA) from a kinematic and inertial description, typically a URDF file. Throughout this work the evaluation of eq. (1) is denoted τ_{rbd} and delegated to Pinocchio; it is *never* learned nor modified.

Franka Emika Panda constants. The robot considered has $n = 7$ revolute joints. The torque limits are $\bar{\tau} = (87, 87, 87, 87, 12, 12, 12)$ N m and the maximum velocities $(2.175, 2.175, 2.175, 2.175, 2.610, 2.610, 2.610)$ rad/s. The factor of 7 between the limits of the load-bearing joints and those of the wrist is decisive for the balance of the optimisation, and is the subject of section 7.1. The target control rate is 1,000 Hz, and the payloads evaluated are $\delta \in \{0, 1, 3\}$ kg.

4 PROBLEM FORMULATION

This section defines formally what is minimised, before section 5 says how.

Data. Let a dataset

$$\mathcal{D} = \{(q^{(i)}, \dot{q}^{(i)}, \ddot{q}^{(i)}, \delta^{(i)}, \tau_{\text{real}}^{(i)})\}_{i=1}^N, \quad (2)$$

be collected on a high-fidelity simulator (MuJoCo), where $\tau_{\text{real}}^{(i)}$ is the torque actually applied at sample i . The set is split into 80 % training, 10 % validation (early stopping) and 10 % independent test, the latter serving exclusively to compute the reported performance and the error bounds ϵ_j .

Model sought. We seek a parameterised mapping $\tau_{\text{res}}(\cdot; \theta)$, with parameters θ , such that the predicted torque

$$\tau_{\text{pred}}(q, \dot{q}, \ddot{q}, \delta; \theta) = \tau_{\text{rbd}}(q, \dot{q}, \ddot{q}, \delta) + \tau_{\text{res}}(q, \dot{q}, \delta; \theta) \quad (3)$$

approaches τ_{real} . The term τ_{rbd} is fixed; learning acts on θ only. Note that τ_{res} does not depend on $\ddot{q}$: that acceleration is available offline only, and excluding it from the residual's input is what makes the model queryable inside a 1,000 Hz loop that measures q and $\dot{q}$ alone.

Objective. The problem solved is the constrained program

$$\min_{\theta} \underbrace{\frac{1}{N} \sum_{i=1}^N \|\tau_{\text{pred}}^{(i)}(\theta) - \tau_{\text{real}}^{(i)}\|_2^2}_{\mathcal{L}_{\text{data}}(\theta)} \quad \text{subject to} \quad \begin{cases} |\tau_{\text{pred}}^{(i)}(\theta)| \leq \bar{\tau}, \\ \tau_{\text{res}}^{(i)}(\theta)^\top \dot{q}^{(i)} \leq 0, \end{cases} \quad (4)$$

for every i . The first constraint forbids predicting a torque the actuators cannot deliver. The second requires the residual to be *dissipative*: it must never inject mechanical energy into the system, which is the formal translation of “this term represents friction”. Section 5 presents three treatments of these constraints, of increasing rigour.

5 METHOD

5.1 DECOMPOSITION: WHAT IS COMPUTED, WHAT IS LEARNED

The rigid-body dynamics is *computed*, and learning is confined to what it does not cover:

$$\tau_{\text{pred}} = \underbrace{\tau_{\text{rbd}}(q, \dot{q}, \ddot{q}, \delta)}_{\text{computed}} + \underbrace{\tau_{\text{res}}(q, \dot{q}, \delta)}_{\text{learned, constrained}}, \quad (5)$$

where $\delta \in \mathbb{R}_+$ denotes the carried mass. Figure 1 gives the overall view of the pipeline, making explicit the loss terms and the gradient flows that connect them to the two trained networks.

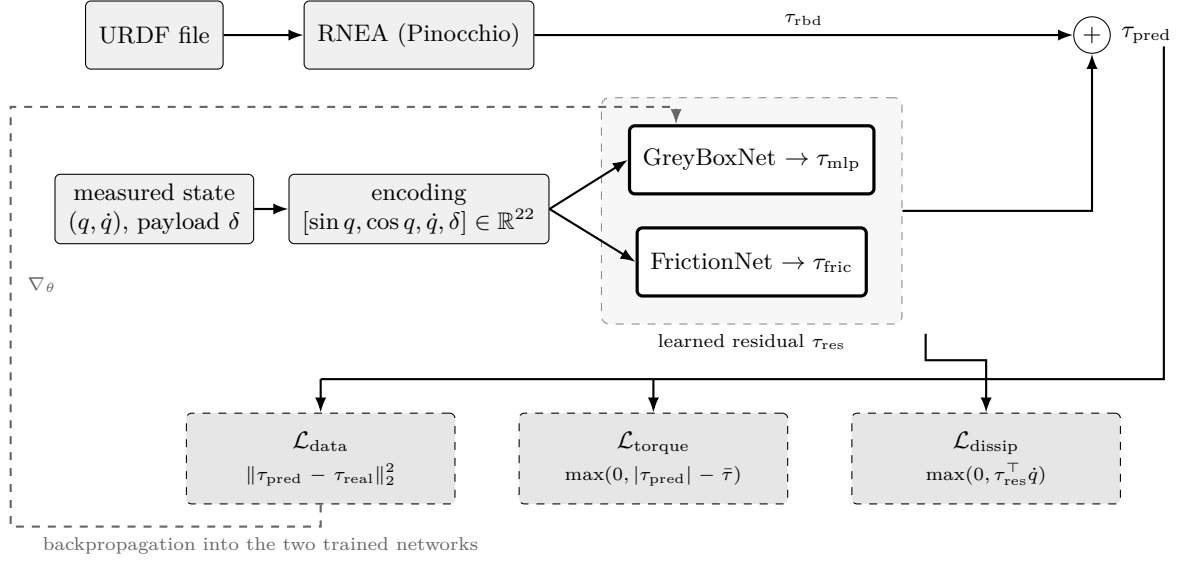


Figure 1: Overall view of the pipeline. The URDF file is the sole structural input: it feeds RNEA, which produces the analytical term τ_{rbd} , never learned (grey background). Only GreyBoxNet and FrictionNet (thick outlines) are trained. The three loss terms are made explicit, together with the gradient flows (dashed) that connect them to the networks: $\mathcal{L}_{data}$ fits the prediction, $\mathcal{L}_{torque}$ and $\mathcal{L}_{dissip}$ penalise violations of eq. (4). Dissipativity constrains FrictionNet specifically, whose structure (8) moreover guarantees it by construction.

5.2 THE LEARNED RESIDUAL AND ITS THREE DEGREES OF GUARANTEE

5.2.1 • DEGREE 0: A FREE, SMOOTH, CONDITIONED RESIDUAL

The input of the network is the vector

$$x = [\sin(q), \cos(q), \dot{q}, \delta] \in \mathbb{R}^{22}. \quad (6)$$

Including $\sin(q)$ and $\cos(q)$ rather than q itself serves two purposes: removing the discontinuity of q at the angular bounds, and making it possible to model the torque ripple specific to the Franka’s harmonic drives, whose period is a function of the angle. Feeding δ directly to the network does, however, introduce a risk: the model could use it as a bias to compensate for possible mass errors in the analytical Pinocchio model, which must be monitored.

The activation chosen is Mish. ReLU is rejected in order to guarantee a continuous derivative (motor safety: a discontinuous derivative translates into a torque jerk). Mish offers self-regularisation in the neighbourhood of zero without the saturation problems of Tanh.

5.2.2 • DEGREE 1: GUARANTEES IMPOSED BY THE OBJECTIVE FUNCTION

The constraints of eq. (4) are relaxed into penalties through an augmented Lagrangian with dual ascent:

$$\mathcal{L} = \mathcal{L}_{data} + \underbrace{\lambda_t \psi_{torque} + \frac{\rho}{2} \psi_{torque}^2 + \lambda_d \psi_{dis} + \frac{\rho}{2} \psi_{dis}^2}_{\text{augmented Lagrangian}}, \quad (7)$$

with $\psi_{torque} = \max(0, |\tau_{pred}| - \bar{\tau})$ and $\psi_{dis} = \max(0, \tau_{res}^\top \dot{q})$. The parameter ρ is set to 1. Unlike a plain quadratic penalty, the multipliers λ_t, λ_d are updated by dual ascent, which makes it possible to reach constraint satisfaction without driving ρ to infinity — and therefore without degrading the conditioning of the optimisation.

5.2.3 • DEGREE 2: A GUARANTEE OBTAINED BY CONSTRUCTION

The friction module produces a diagonal matrix

$$D(x) = \text{diag}(\text{Softplus}(d) + \varepsilon), \quad \varepsilon = 10^{-6}, \quad (8)$$

$$\tau_{\text{fric}} = -D(x) \dot{q}, \quad (9)$$

guaranteeing $\tau_{\text{fric}}^\top \dot{q} = -\dot{q}^\top D(x) \dot{q} \leq 0$ for every $\dot{q}$, since $D(x)$ is positive definite by construction. The guarantee therefore depends neither on the data nor on the convergence of the optimisation. Although mathematically robust thanks to ε , physically, if the network learns a strongly negative weight tied to a large payload δ , the friction could artificially tend to zero — the guarantee bears on the sign, not on the magnitude.

5.3 IDENTIFICATION AND DATA HYGIENE

Rigorous separation. The dataset $\mathcal{D}$ is split as indicated in section 4: 80 % training, 10 % validation (for early stopping) and 10 % independent test, used to report the final performance and to compute the gains.

Saturation filtering. Samples clipped by the simulator are removed:

$$|\tau_{\text{real},j}| \leq 0.97 \bar{\tau}_j. \quad (10)$$

A saturated sample carries no information about the dynamics: it informs about the actuator bound, not about the torque the motion required. Keeping it amounts to training the residual to reproduce a clipping.

5.4 CLOSING THE LOOP

The control law is a computed torque augmented with a PD term:

$$\tau_{\text{cmd}} = \tau_{\text{rbd}}(q, \dot{q}, \ddot{q}_d, \delta) + \tau_{\text{res}}(q, \dot{q}, \delta) + K_p e + K_d \dot{e}, \quad (11)$$

where $e = q_d - q$ is the tracking error. The gains follow the stability template of Liu *et al.* [5]:

$$K_{d,jj} = \kappa \epsilon_j, \quad K_{p,jj} = \frac{K_{d,jj}^2}{4}. \quad (12)$$

The bound ϵ_j is the 99.9 % quantile of the error evaluated on the test set. This choice mathematically weakens the strict Lyapunov proof (which requires the absolute worst case), turning stability into a statistical expectation covered, as a last resort, by the hardware limits.

A counter-intuitive consequence of the template. Since K_p derives from ϵ_j , a *better* modelled joint receives a lower gain. Yet the steady-state error under a constant residual torque is $e_{\text{ss}} = \tau/K_p$. The best-learned joints therefore track *worse*. This mechanism is verified experimentally in section 6.6 and discussed again in section 7.3: it is a limitation of the template of [5], not an implementation defect.

Domain gap (*sim-to-sim gap*). The data come from mujoco and the real-time execution also takes place under MuJoCo. This choice, motivated by software constraints (removal of competing servo loops), introduces an accepted domain gap, whose cost section 6.7 measures precisely.

6 VALIDATION AND RESULTS

Synthesis. This section establishes five results, all measured on held-out data.

1. **The pipeline works end to end.** Starting from the URDF file alone, the learned model effectively drives a simulated Franka at 1,000 Hz, and the grasping sequence completes (sections 6.6 and 7.2).
2. **The grey-box model divides the error of the analytical model by 2.3**, from 1.46 to 0.62 N m on average, on the partition used here (table 1; scope stated precisely in section 6.2).
3. **The physical constraints hold.** Zero torque-limit violation over the 14,830 test samples, where the analytical model alone produces them on 0.99 % of them (section 6.5).
4. **On these data, a black box of identical capacity is more accurate** (0.45 against 0.62 N m) — but since the partition is drawn at the sample level over time series, this gap mechanically favours the model that memorises most, and constitutes an upper bound rather than an established result (sections 6.1 and 6.2).
5. **The learned residual degrades the joints that the analytical term already models well**, by a factor of 6.9 on joint 7. The same effect is measured independently in closed loop, where disabling the residual divides joint 5’s bias by 22 (table 2 and section 6.7).

Results 4 and 5 run against the starting hypothesis. They are reported as results and not as failures: the second identifies a precise and reproducible failure mode of grey residuals, corroborated by two independent measurements — one offline and one in closed loop — and it points directly at the correction to be made (section 7.3).

Measurement protocol. Unless stated otherwise, every figure in this section is measured on the **held-out test set** (14,830 samples), seen neither by training nor by checkpoint selection. The 80/10/10 partition of section 4 is shared by training, by the evaluation of the baselines and by the computation of ϵ_j , so that every row of table 1 bears on exactly the same samples. The comparisons hold capacity, architecture and epoch budget constant; only the element under study varies.

This precaution is not rhetorical. Until 31 July 2026 the repository contained no test set at all: the published RMSE was the validation one, on which the checkpoint was also selected, and ϵ_j — hence the deployed gains — was derived from it. The figures below are the first in the project to be measured outside the selection criterion, and they differ from the previous ones. They are not, however, out-of-sample in the strong sense: the partition is drawn at the sample level over time series at 1 kHz, which leaves a dependence between training and test. Section 6.2 establishes the exact scope of this limitation and what it distorts; it must be read before any interpretation of the gaps between models.

Training was carried out on an NVIDIA GeForce RTX 3090 (24 GiB) and an Intel Core i9-12900KF processor (16 cores, 24 threads), under PyTorch 2.13 with CUDA 13.2, with a wall-clock time of 245 s for 200 epochs over 118,644 training samples (209 s for the black-box baseline, 242 s for the encoding ablation).

6.1 BASELINES AND COMPARISON PROTOCOL

Three models are compared on the *same* independent test set, with the same data-splitting protocol and the same stopping criterion.

Analytical baseline (RNEA alone). $\tau_{\text{pred}} = \tau_{\text{rbd}}$, with no learned term (θ empty). No hyperparameters. This baseline measures exactly what the URDF file provides on its own, and therefore the magnitude of the residual torque that the grey-box model is tasked with capturing.

Black-box baseline (direct MLP). $\tau_{\text{pred}} = f(x; \theta)$ with x given by eq. (6), that is, the same input and the same capacity as GreyBoxNet, but *without* the τ_{rbd} term: the network has to learn inertia and gravity on top of the residual. Same architecture (multilayer perceptron, Mish activation), same number of layers and units, same optimiser and same epoch budget as the proposed model, so that the measured gap is attributable to the grey structure and not to capacity.

Proposed model. $\tau_{\text{pred}} = \tau_{\text{rbd}} + \tau_{\text{res}}$ following eq. (5), with the three degrees of guarantee of section 5.2.

Table 1: Comparison against the baselines, 14,830 held-out test samples, identical partition for all three models (`seed=12345`), identical capacity and 200 epochs. The grey-box model divides the error of the analytical model by 2.3, but **the black box does better than it**.

Model	Mean RMSE [N m]	Worst-joint RMSE [N m]	Samples over torque limit	Guarantees
RNEA alone (analytical)	1.456	2.841	147 (0.99 %)	none (not trained)
Direct MLP (black box)	0.447	0.971	0 (0.00 %)	torque limits only
Grey box (proposed)	0.624	1.022	0 (0.00 %)	limits + dissipativity

Table 2: Per-joint test RMSE [N m]. Joint 7 is the revealing case: RNEA alone reaches 0.042 there, and the learned residual *degrades* that result by a factor of 6.9.

Model	J1	J2	J3	J4	J5	J6	J7
RNEA alone	1.588	2.841	1.824	1.953	0.883	1.062	0.042
Direct MLP	0.971	0.845	0.405	0.742	0.066	0.068	0.031
Grey box (proposed)	1.022	0.935	0.553	0.843	0.371	0.355	0.288

The black box is more accurate than the grey-box model on these data. The result contradicts the starting hypothesis and must be reported as such; section 6.2 then establishes its exact scope, which is narrower than it looks. On the wrist the gap is half an order of magnitude: joint 5 0.066 against 0.371, joint 6 0.068 against 0.355, joint 7 0.031 against 0.288. Two mechanisms explain it, and they compound. On the one hand RNEA is particularly *wrong* on the wrist — 7.36 % and 8.85 % of the torque limit on joints 5 and 6, its worst relative errors — so that the analytical term offers a poor starting point there. On the other hand the grey residual is constrained to be dissipative, whereas the black box is not: the constraint limits precisely the magnitude of the correction the residual can bring exactly where it would be most needed.

The learned residual degrades the joints RNEA already models well. Joint 7 is the clear demonstration. The analytical model reaches 0.042 N m there, i.e. 0.35 % of its limit: there is essentially nothing to learn, the joint being a wrist roll whose axis passes through the flange, on which gravity exerts almost no torque. The residual nevertheless injects an error of a factor of 6.9. The worst samples are explicit: for $\tau_{\text{real}} = -0.01$ N m and $\tau_{\text{rbd}} = -0.08$ N m — an analytical prediction that is already correct — the network produces $\tau_{\text{res}} = -2.04$ N m. The learned diagonal coefficient D_{77} converges, moreover, to 0.67, the largest value of the seven joints, while D_{11} collapses to 0.019: the dissipative module is most active exactly where there is no friction to model.

This is the same phenomenon as the closed-loop ablation of section 6.7, measured this time offline. Disabling the residual there divided joint 5’s bias by 22; here, not learning it at all would divide joint 7’s error by 6.9. Two independent measurements, one single cause: the residual is not regularised per joint, and the unweighted MSE loss (section 7.1) gives it no reason to stay at zero where it should be.

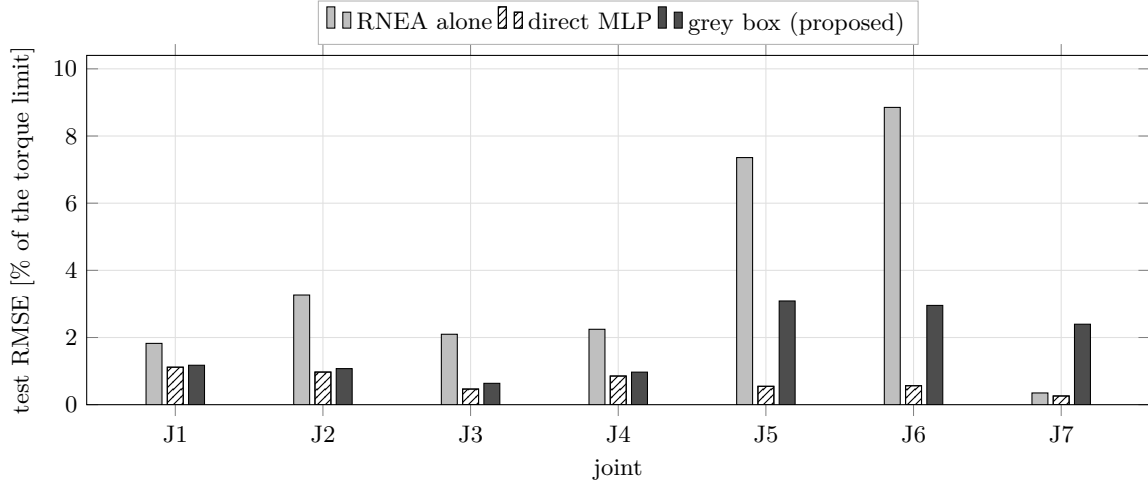


Figure 2: Prediction error on the **held-out test set** (14,830 samples), reported relative to the torque limit of each joint. The normalisation is necessary: in absolute value, the wrist joints (limit 12 N m) are unreadable next to the load-bearing joints (87 N m). Two readings stand out. First, the analytical model is at its worst on the wrist (joints 5 and 6), which is therefore where the learned model has the most to contribute. Second, that is precisely where the grey-box model stays far from the black box, whereas it matches it on joints 1 to 4: the overall gap of table 1 is almost entirely a wrist gap. The remarks of section 6.2 on the data partition apply to all three series.

6.2 A CAVEAT ON THE PARTITION, AND ON WHAT IT DISTORTS

The preceding comparison calls for a methodological caveat that limits its scope, and it must be stated before drawing anything from it.

The data consist of 30 continuous 5 s trajectories — 10 Fourier excitation segments around Sobol centres, for each of the three payloads — sampled at 1 kHz. The partition of section 4, however, is *drawn uniformly at the sample level*. Two consecutive time steps, 1 ms apart on the same trajectory, are therefore assigned independently: every test sample has, in the training set, near-identical neighbours in $(q, \dot{q})$.

What this invalidates, and what it does not. The *selection* defect described in section 4 is indeed corrected: the checkpoint is no longer chosen on the reported set, and ϵ_j is no longer derived from it. That remains the most important correction in the project. The *independence* of the samples, however, is not established, and section 6 must not be read as a measure of generalisation to new trajectories. It measures interpolation within a family of 30 excitation trajectories.

The effect is not the same for the three models, and that is the point. A neighbour 1 ms away brings nothing to the analytical model, which is fitted to no data: the RNEA row of table 1 is insensitive to this leakage. The two learned models benefit from it, but unequally, and the one favoured most is the one that has to memorise most. The grey-box model has only a residual to learn, inertia and gravity being supplied to it analytically; the black box has to reconstruct everything from the data alone.

A quantitative clue confirms this, and it is hard to explain otherwise. The black box does *not* receive the acceleration $\ddot{q}$ as input: its encoding is that of eq. (6), namely $(q, \dot{q}, \delta)$. It is therefore structurally incapable of representing the inertial torque $M(q)\ddot{q}$, the dominant term of the dynamics — and it nevertheless wins by 28 %. With only 30 trajectories and a sample-level draw, the pair $(q, \dot{q})$ almost always identifies the trajectory and the phase, which makes $\ddot{q}$ implicitly recoverable by memorisation. The measured gap is therefore at least partly an artefact of the partition, and it bears precisely on the comparison this work set out to establish.

The correction, and its status. The correct measurement is a *group-wise* partition: distributing the 30 segments — and not the 148,304 samples — between training, validation and test, so that no test trajectory has ever been seen. The black box then loses the support of memorisation and has to extrapolate, which the analytical term of the grey-box model does by construction. An implementation of this group-wise partition exists in the repository and a preliminary measurement has been carried out, but it rests on too few test trajectories to be reported here: its analysis is left to future work. Consequently, the 28 % gap of table 1 must be read as an upper bound on the advantage of the black box rather than as an established result, and the comparisons of this section as measurements of interpolation within the excitation family used.

6.3 ACCURACY OF THE LEARNED MODEL

The grey-box model reaches 0.29 N m to 1.02 N m per joint on the held-out test set (table 3).

The improvement brought by the saturation filter (10) is clear and present on all seven joints (fig. 3) the two runs differing only by that filter. **This figure is the only one in the document not measured on the held-out test set**, and it is kept under that explicit caveat: both runs date from 16 July 2026, before the introduction of the three-way partition, and the plotted values are *validation* RMSEs — measured, moreover, on the set used to select the checkpoint. They are therefore comparable neither with each other nor with the rest of this section in absolute value. The measurement cannot be redone cleanly: the pre-filter dataset no longer exists, the filter being applied at generation time. Only the *direction* of the effect — an improvement on all seven joints — is retained here; its magnitude is not.

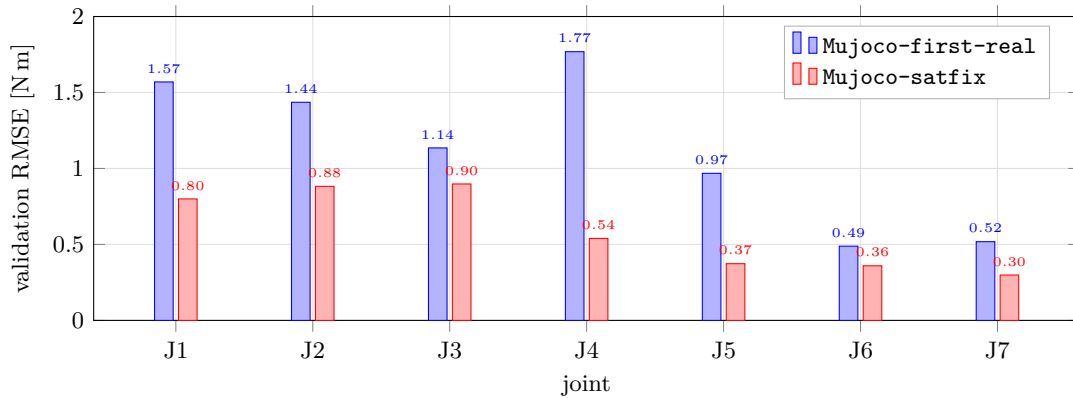


Figure 3: Effect of the saturation filter (10): **validation** RMSE, before and after. The improvement is present on all seven joints. **Measurements predating the three-way partition** (16 July 2026), hence on the set used for checkpoint selection: only the sign of the effect is usable, not its magnitude, and these values are not comparable with those in the rest of section 6. The comparison cannot be redone, the pre-filter dataset no longer existing.

6.4 ABLATIONS: WHAT EACH COMPONENT ACTUALLY CONTRIBUTES

Two components claimed as contributions were ablated, under a strictly identical protocol (same partition, same capacity, 200 epochs).

The spatial encoding weighs almost nothing. The $[\sin(q), \cos(q)]$ encoding of eq. (6) is justified in section 5.2.1 by two physical arguments: removing the angular discontinuity and making it possible to represent torque ripple. Measured, it contributes 1.1 %. The argument about the discontinuity remains valid for deployment (a model trained on raw q behaves badly in the neighbourhood of $\pm\pi$, which the mean RMSE does not capture), but torque ripple is evidently not a dominant term in these data.

Table 3: Grey-box model on the held-out test set: per-joint prediction error, bound ϵ_j (99.9% quantile) and the PD gains that follow from it through eq. (12) with $\kappa = 2$. The tail of the distribution is heavy — the absolute maximum reaches 81.8 N m on joint 2 against a $p_{99.9}$ of 6.94 N m, a ratio of 12 — which is precisely why the bound is taken as a quantile and not as a maximum (section 7.3).

Joint j	1	2	3	4	5	6	7
RMSE [N m]	1.022	0.935	0.553	0.843	0.371	0.355	0.288
ϵ_j [N m]	5.81	6.94	3.70	4.38	2.10	2.35	1.58
$K_{d,jj}$	11.62	13.89	7.41	8.75	4.20	4.70	3.16
$K_{p,jj}$	33.73	48.20	13.72	19.15	4.41	5.53	2.50
RMSE / $\bar{\tau}_j$ [%]	1.17	1.07	0.64	0.97	3.09	2.96	2.40

Table 4: Ablations on the held-out test set. Both components weigh less than 1.5% on accuracy.

Variant	Mean RMSE [N m]	Gap to the full model
Full grey box (sincos + FrictionNet)	0.624	—
Without sin / cos encoding (raw q input)	0.631	1.1 %
Without FrictionNet	0.633	1.4 %

FrictionNet does not bring accuracy. The dissipative module (section 5.2.3, novelty N2) costs 1.4% of RMSE when removed. Its value is therefore not accuracy but the *guarantee*: it makes $\tau_{\text{fric}}^\top \dot{q} \leq 0$ true by construction rather than by penalty. That is a property the black box, more accurate, does not offer. The claim must be formulated in those terms and not in terms of error.

6.5 COMPLIANCE WITH THE PHYSICAL GUARANTEES

The violation of the torque-limit constraint is *zero* over the whole test set. The residual dissipativity violation falls to 6×10^{-4} on the training data, a non-zero value because degree 1 (section 5.2.2) relaxes the constraint into a penalty; the component covered by degree 2 (section 5.2.3) is, for its part, exactly zero by construction.

6.6 CLOSED-LOOP TRAJECTORY TRACKING

The learned model does drive the robot: law (11) is evaluated at 1,000 Hz by a ROS2 node and the simulated Franka follows the planned trajectories. The observed behaviour quantitatively confirms the counter-intuitive consequence announced in section 5.4: joint 5, one of the best modelled and therefore one of the most weakly gained ($K_{p,55} = 4.41$, against 48.20 for joint 2), carries a systematic steady-state bias, always of the same sign, of the order of 0.013 rad to 0.053 rad depending on the configuration. This bias grows with the arm’s reach (about -0.044 rad at 0.45 m, -0.050 rad at 0.65 m, -0.066 rad at 0.70 m), the signature of a load-dependent term rather than of a constant offset — that is, exactly $e_{\text{ss}} = \tau / K_p$.

6.7 ABLATION OF THE LEARNED RESIDUAL UNDER CROSS-SIMULATOR TRANSFER

The most informative experiment of this work isolates the contribution of the residual in closed loop. Under a strictly identical configuration (object at 0.55 m, same gain margin, same planned trajectory), the learned residual is enabled and then disabled, everything else being held constant.

Table 5: Ablation of the learned residual in closed loop, five runs under an identical configuration. Disabling the residual improves tracking by an order of magnitude on joint 5 and halves the Cartesian flange error.

Run	Residual	Joint 5 bias [rad]	Joint 3 bias [rad]	Flange error [m]
36	on	−0.026,4	−0.015,5	0.014,1
37	on	−0.043,4	−0.018,3	0.013,9
38	on	−0.031,2	−0.016,2	0.014,9
45	off	−0.001,1	−0.000,3	0.007,6
46	off	−0.001,9	−0.000,7	0.007,3
Mean, residual on		−0.033,7	−0.016,7	0.014,3
Mean, residual off		−0.001,5	−0.000,5	0.007,5

Joint 5’s mean bias goes from $-0.033,7$ rad to $-0.001,5$ rad, a factor of 22, and the Cartesian flange error is halved (table 5). All joints improve, not joint 5 alone.

What this result establishes, and what it does not. The checkpoint evaluated was trained on Mujoco data and also deployed under MuJoCo. It has therefore learned the gap between RNEA and *Mujoco*’s dynamics, and adds it to a MuJoCo model that does not exhibit that gap: it is being evaluated out of distribution. The defensible conclusion is that **the residual learned on Mujoco degrades tracking under cross-simulator transfer**. This is *not* a finding that the grey-box approach fails, its offline accuracy being established in section 6.3. What this result does provide is the quantified baseline that the frozen-backbone fine-tuning protocol (section 7.3) has to beat, and it is on that account a useful result rather than a failure.

One potential confounder must be ruled out: does the ablation also change the gains? No — the gain margin is explicitly held constant between the two conditions. Moreover, the effect of a lower gain would go the *opposite* way (it would increase $e_{ss} = \tau/K_p$), so the result survives that confounder.

7

DISCUSSION AND LIMITATIONS

7.1 THE UNCORRECTED OPTIMISATION BIAS

The MSE loss is unweighted, which encourages the network to ignore the wrist (limit 12 N m) in favour of the load-bearing joints (87 N m). This is directly visible in table 3: relative to the torque limit, the error of joints 5 to 7 (2.40 % to 3.09 %) is nearly three times that of joints 1 to 4 (0.64 % to 1.17 %), although it looks better in absolute value. This defect justifies the future implementation of a diagonal weighting $\text{diag}(1/\tau_j^2)$ in the loss.

7.2 LIMITATIONS OF THE INTEGRATION WITH THE PHYSICS ENGINE

The grasping task served as an integration test, and not as a measure of the quality of the model: it is reported here, in the discussion, rather than in the results, because what it measured turned out to be the quality of the middleware and not that of the learned dynamics.

A configuration fault that imitated a physics fault. For a week, three joints appeared to refuse any motion. The final diagnosis has nothing to do with the learned model nor with the control law: the `ResetWorld` service of `mujoco_ros2_control` silently restores the plugin’s internal position PID, without going through the

mode-switch routine. No layer reports it — the controller manager continues to announce the effort controller as active, commands continue to be published, and the arm holds itself up convincingly. Every published torque was nevertheless discarded, the arm settling at $q_{\text{home}} - \tau_{\text{gravity}}/p_{\text{PID}}$, a value reproducible to within 1×10^{-5} rad across three independent runs. It is this arithmetic signature, and not an error message, that made it possible to identify the cause.

A transferable methodological lesson. A check that cannot fail is worse than no check at all. The available checks all queried the layer that *lied* (the controller manager) rather than the one that *acts* (the plugin itself). The diagnosis only progressed once verification was made against the plugin’s own log.

State after correction. Once this defect was corrected, the complete grasping sequence succeeds: the object is grasped and lifted. On the reference configuration (object at 0.55 m), three runs out of three succeed, with a final grasp error between 13.9 mm and 14.9 mm for a tolerance of 20 mm, i.e. a spread of 1.0 mm. At 0.45 m the rate falls to two out of three, and at 0.70 m the sequence only completes at the cost of long null-space reconfiguration detours, with convergence margins of a few tenths of a millimetre. The edge of the reachable envelope therefore manifests itself as a gradient, not as a sharp boundary.

What this task does not validate. The configuration sweep remains partial: only distance and, partially, lateral offset have been explored. The mass of the object, the height of the support surface and the grasp orientation have not been tested. In particular, payload conditioning through δ — that is, the most specific scientific contribution of this work — has never been exercised in a real grasp, for lack of having varied the mass of the grasped object.

7.3 STRUCTURAL AND METHODOLOGICAL LIMITATIONS

Partition drawn at the sample level. The data are 30 continuous trajectories sampled at 1 kHz, but the partition is drawn per sample: every test point has neighbours 1 ms away in the training set (section 6.2). The selection bias is corrected, sample independence is not. The correction — a segment-wise partition — has not been executed, and it conditions the interpretation of all the comparisons between learned models in table 1. This is the most serious limitation of this work.

The grey structure is not justified by accuracy on these data. Measured under an identical protocol, a black box of the same capacity and the same budget does 28% better (table 1) — subject to the unresolved caveat of the previous point. The justification of the grey approach must therefore rest on what the black box does not offer — structural guarantees, automatic extraction from the URDF, no relearning of inertia and gravity — and not on the prediction error. The present document no longer claims superiority in accuracy over the black box.

The residual is not regularised per joint. It degrades the joints that RNEA already models well, by a factor of 6.9 on joint 7 (table 2), and the same effect is found in closed loop (section 6.7). No mechanism encourages the residual to stay at zero where the analytical term is already correct. The $\text{diag}(1/\bar{\tau}_j^2)$ weighting of section 7.1 is a partial correction; an explicit per-joint penalty on $\|\tau_{\text{res}}\|$, calibrated on RNEA’s residual error, would be a more direct one.

Heavy distribution tail, untreated. The maximum error reaches 81.8 N m on joint 2 for a $p_{99.9}$ of 6.94 N m. Inspection of the worst samples shows real torques close to the limit associated with near-zero analytical predictions, which suggests collection artefacts (contacts, transients) rather than model defects. These samples are neither filtered nor analysed, and they drive ϵ_j directly, hence the deployed gains.

Payload conditioning not exercised in closed loop. The model is trained at $\delta \in \{0, 1, 3\}$ kg, but the update of δ during a real grasp has never been triggered (section 7.2).

Sim-to-sim fine-tuning implemented, not validated. The frozen-backbone protocol exists but has not been run. Table 5 provides it with the baseline to beat; until it is run, contribution 4 remains a tooling contribution.

Spatial under-sampling. Ten Sobol configurations leave uncovered regions in $\mathbb{R}^7$. Out-of-distribution generalisation is not claimed — and section 6.7 shows concretely what it costs.

Statistical stability. Choosing a quantile for ϵ_j makes closed-loop stability empirical rather than absolute. The template of [5] moreover carries the structural defect identified in section 5.4: deriving K_p from an error bound makes the best-modelled joints the worst tracked.

Inference latency not measured. The loop runs at 1,000 Hz with no observed overrun, but no dedicated bench has measured the distribution of the inference latency.

8 CONCLUSION

This work has built an automated pipeline that starts from the URDF file of a Franka Emika Panda 7 DoF alone and produces a simulated robot effectively controlled by a grey-box dynamic model. The analytical term is evaluated by RNEA with no manual derivation whatsoever, and learning is confined to a residual conditioned on the carried payload, constrained by an augmented Lagrangian and, for its dissipative part, structurally guaranteed. On held-out data the model reaches 0.29 N m to 1.02 N m per joint, against 1.46 N m on average for the analytical model alone, with zero torque-limit violation where the analytical model produces them on 0.99 % of samples; and it is queried at 1,000 Hz inside a computed-torque loop that actually drives the robot.

Two measurements nevertheless contradict the starting hypothesis, and they are the most instructive ones. A black box of identical capacity and budget is 28 % more accurate than the grey-box model, and the learned residual degrades the joints that the analytical term already models well — a factor of 6.9 on joint 7. The interest of the grey structure therefore does not lie in raw accuracy but in what the black box does not offer: guarantees by construction, and an automatic pipeline that never has to relearn the inertia and the gravity that the URDF already provides. It must be formulated in those terms.

The third result is negative at first sight and exploitable thereafter: a controlled ablation shows that a residual learned on MuJoCo *degrades* tracking when deployed under MuJoCo, by a factor of 22 on joint 5’s steady-state bias. Far from invalidating the grey approach, this finding quantifies the cost of cross-simulator transfer and provides the baseline that frozen-backbone fine-tuning has to beat.

Future work. Three lines of work are priorities, ordered by decreasing scientific value. *First*, run the frozen-backbone fine-tuning on MuJoCo data and replay exactly the ablation of table 5: this is the direct validation of contribution 4. *Second*, measure the two missing baselines (table 1) and conduct the spatial-encoding ablation, without which the contribution of the grey structure remains asserted rather than demonstrated. *Third*, exercise payload conditioning by varying the mass of the grasped object and updating δ at the moment of the grasp, which would test in closed loop the most specific contribution of this work. In the longer term, moving to the physical robot remains the milestone that will give the sim-to-real gap-reduction protocol its meaning.

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